Problem Set 4

AI 501: Mathematics for Artificial Intelligence | Fall 2026

Problem 1

Let

$$A = \begin{pmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{pmatrix}.$$

(a) Reduce A to echelon form. Find $\operatorname{rank}(A)$, the pivot and free variables, and a basis for $\mathcal{N}(A)$.

(b) Find the condition on $b = (b_1, b_2, b_3)^T$ for $Ax = b$ to be solvable.

(c) For $b = (1, 5, 6)^T$, find the complete solution $x = x_p + x_n$.

(d) The lecture slide says "if $\operatorname{rank}(A) < n$, the system has infinitely many solutions." Use this example to state the claim correctly.

Problem 2

Let A be $m \times n$ with rank r. For each case, state how many solutions $Ax = b$ can have, and give a small example matrix.

(i) $r = m = n$

(ii) $r = n < m$

(iii) $r = m < n$

(iv) $r < m$ and $r < n$

Problem 3

Construct a matrix with each property, or explain why it is impossible.

(a) Its column space contains $(1, 1, 0)^T$ and $(0, 0, 1)^T$, and its row space contains $(1, 2)^T$ and $(2, 5)^T$.

(b) Its column space has basis $\{(1, 1, 3)^T\}$, and its null space has basis $\{(3, 1, 1)^T\}$.

(c) $\dim \mathcal{N}(A) = 1 + \dim \mathcal{N}(A^T)$.

Problem 4

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$.

(a) Show that $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$ and $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$.

(b) Hence prove $\operatorname{rank}(AB) \leq \min(\operatorname{rank} A, \operatorname{rank} B)$.

(c) Give 2×2 matrices for which the inequality is strict.

Problem 5

Suppose every label is predicted by a single constant c, given data $y_1, \ldots, y_n$.

(a) Show that the c minimizing $\frac{1}{n}\sum_i (y_i - c)^2$ is the sample mean.

(b) Show that the c minimizing $\frac{1}{n}\sum_i |y_i - c|$ is a sample median.

(c) For $y = \{2, 3, 4, 5, 100\}$, find both minimizers. Use the result to explain the lecture's remark that MAE is suited for noisy data.

Problem 6

The training data are:

Obs	X_1	X_2	X_3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	−1	0	1	Green
6	1	1	1	Red

We want to predict Y at the test point $X_1 = X_2 = X_3 = 0$.

(a) Compute the Euclidean distance from each observation to the test point.

(b) What is the prediction with $k = 1$? Why?

(c) What is the prediction with $k = 3$? Why?

(d) If the class decision boundary in this problem is highly nonlinear, would you expect the best k to be large or small? Why?

Problem 7

Consider brute-force kNN with n training points in $\mathbb{R}^d$.

(a) Give the time complexity of classifying one test point, broken down into computing distances, selecting the k nearest, and voting. Also give the storage cost. For MNIST (n = 60,000, $d = 784$), roughly how many multiply-adds does one query need?

(b) Why is an odd k recommended for binary classification? Does an odd k prevent ties when there are three classes?

(c) A dataset records each person's annual salary in dollars and age in years. When kNN uses Euclidean distance, a difference of \$1,000 in salary counts far more than a difference of 50 years in age. Explain why, and give the fix.

Problem 8

A screening test on 2030 people gives $TP = 20$, $FP = 180$, $FN = 10$, $TN = 1820$.

(a) Compute sensitivity, specificity, FPR, precision, negative predictive value NPV $= \frac{TN}{TN+FN}$, and accuracy.

(b) The test has 91% specificity, yet only 10% of positive results are true positives. Explain why.

Problem 9

The Matthews correlation coefficient is

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}.$$

(a) A dataset has 91 positives and 9 negatives. A classifier gives $TP = 90$, $FN = 1$, $TN = 0$, $FP = 9$. Compute accuracy, F_1, and MCC. Which metric tells the truth about this classifier?

(b) Compute F_1 and MCC for the lecture's heart-disease example: $TP = 100$, $FP = 10$, $FN = 5$, $TN = 50$.

Problem 10

Twenty test instances sorted by classifier score (p = actually positive, n = actually negative):

0.9	0.8	0.7	0.6	0.55	0.54	0.53	0.52	0.51	0.505
p	p	n	p	p	p	n	n	p	n
0.4	0.39	0.38	0.37	0.36	0.35	0.34	0.33	0.30	0.1
p	n	p	n	n	n	p	n	p	n

(a) An instance is predicted positive if its score is at least t. Give the (FPR, TPR) point for $t = 0.54$, 0.51, and 0.38.

(b) Compute the accuracy at $t = 0.54$.

(c) Compute the AUC, using the interpretation that AUC is the probability that a randomly chosen positive is scored higher than a randomly chosen negative.

(d) Where do the perfect classifier and a random classifier lie in ROC space?